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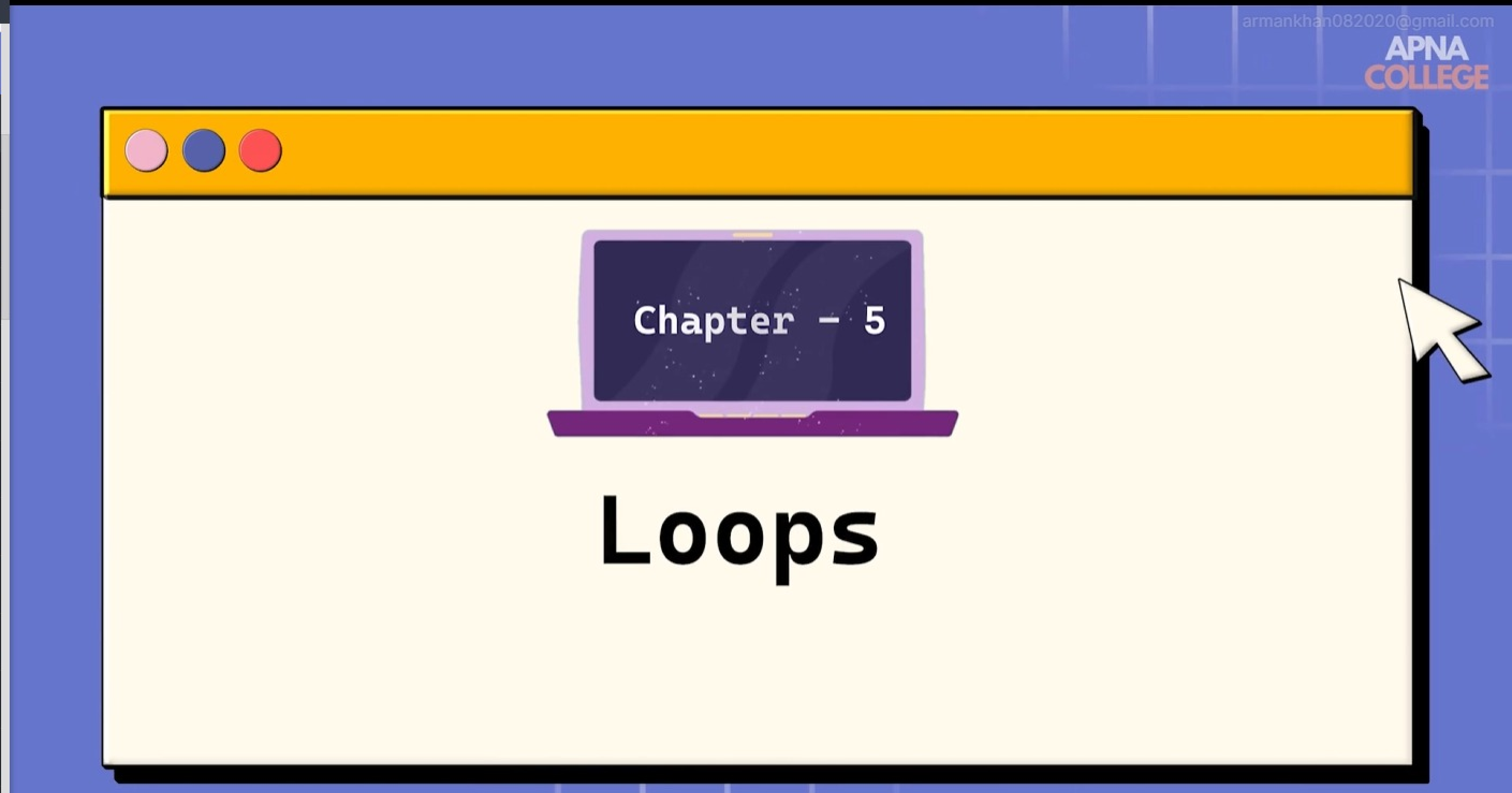
Sigma 9.0 (C++ DSA)

6 %

10. Loops



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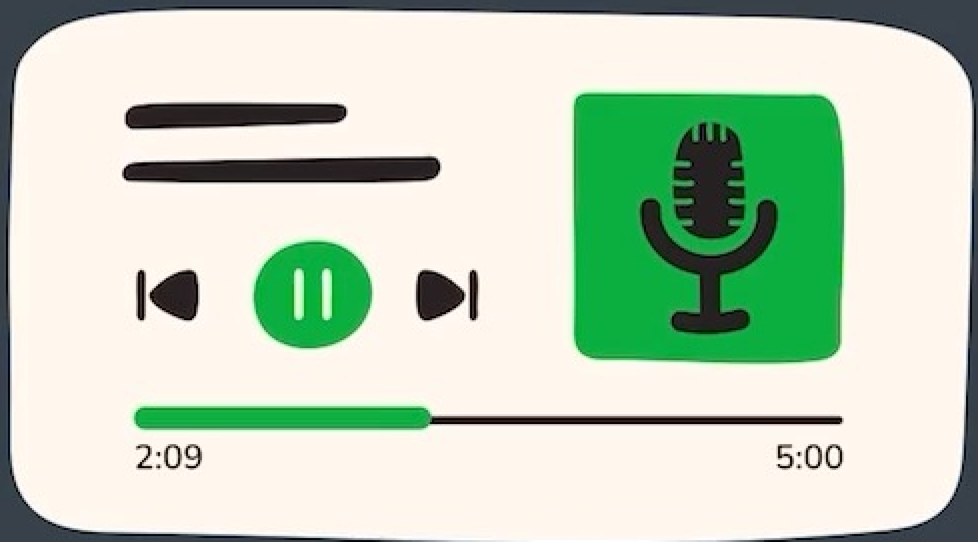
Loops

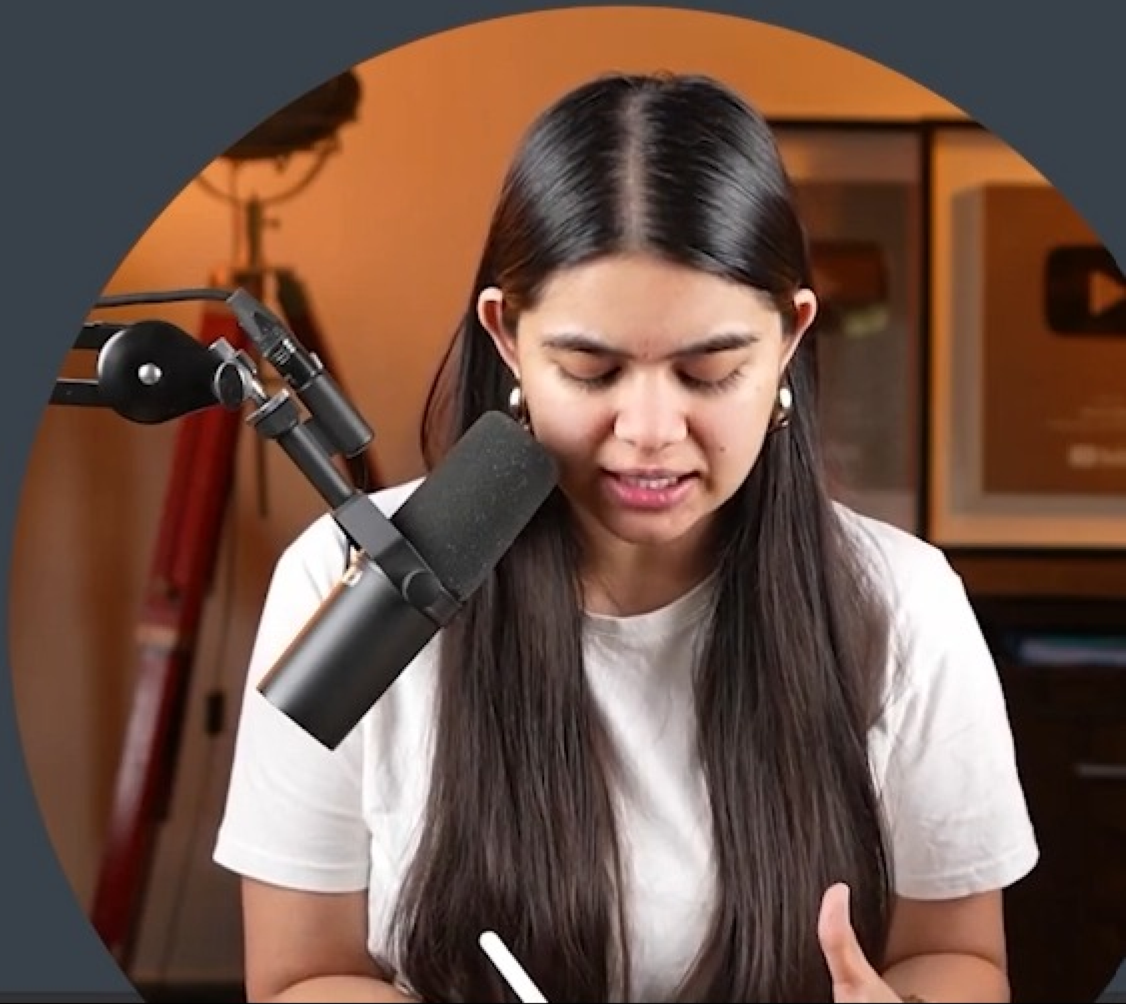


- for Loop
- while Loop
- do-while Loop

for Loop

Loop runs while
condition is true

```
for(initialisation; condition; updation) {

    //do some work

}
```



for Loop

Loop runs while condition is true

```
for(initialisation; condition; updation) {

    //do some work

}
```



```
int num = 1;    //st 1

num <= 5        //condition (st 2)

num++           // st 3
```

```
for( int num = 1; num <= 5; num++) {
    cout << num;
}
```

for Loop

Loop runs while

condition is true

```
for(initialisation; condition; updation) {

    //do some work

}
```

```
for( int num = 1; num <= 5; num++) {
    cout << num;
}
```

DRY Run

num = 1

num = 2

num = 3

num = 4

num = 5

num = 6

1

2

3

4

5

for Loop

infinite ∞
Loop

Loop runs while
condition is true

```
for(initialisation; condition; updation) {

    //do some work

}
```

true

iterate (run)

iteration (single run)

for Loop

Loop runs while condition is true

```
for(initialisation; condition; updation) {

    //do some work

}
```

Print "apna college" 5 times

Print "apna college" 5 times

Dry Run

```
for(int i=1; i<=5; i++) {
    cout << "apna college" << endl;
}
```


APNA COLLEGE

Print numbers from 1 to n

$n = 15, 20, 30$ (user)

Cpp Codes

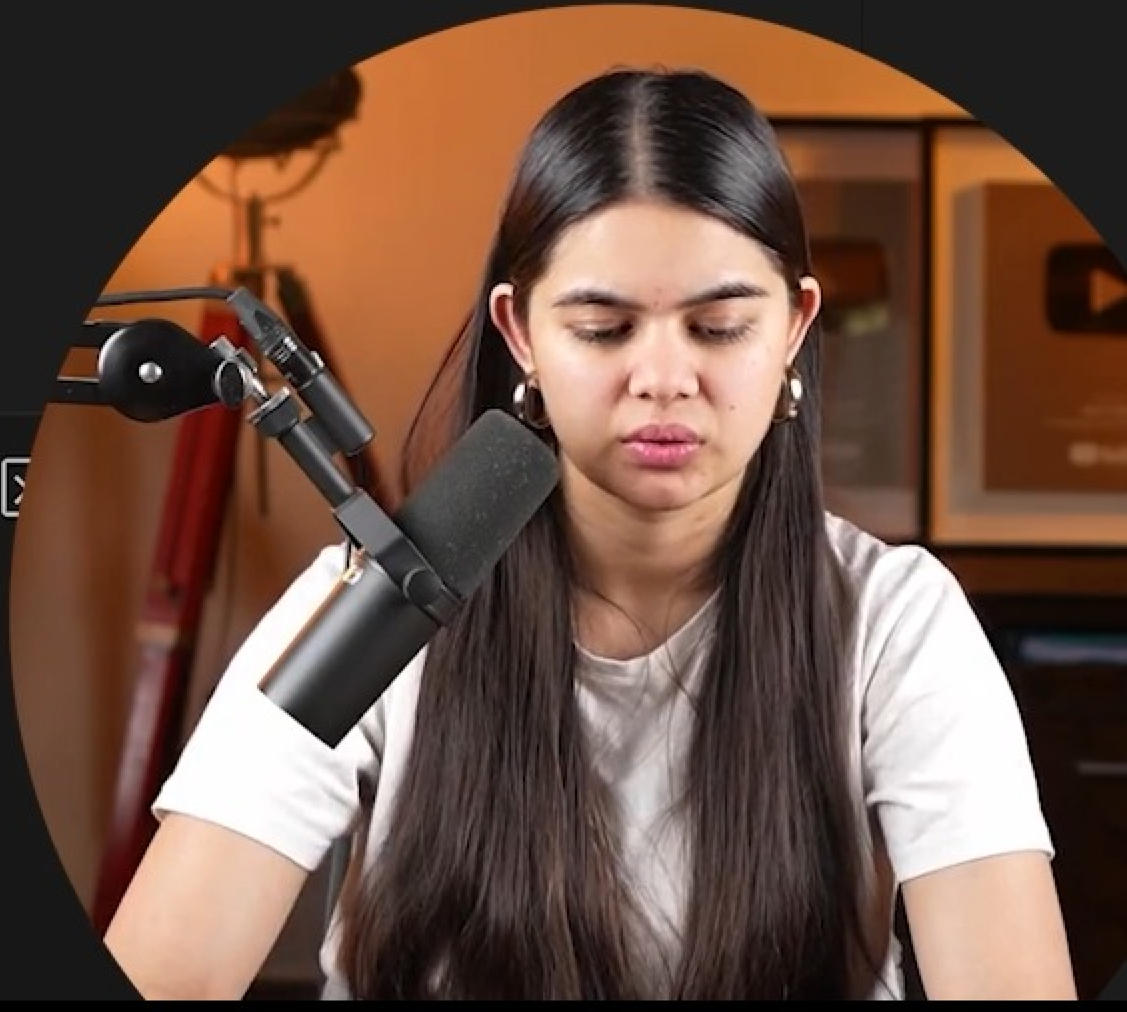
code.cpp

code.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "enter your n : ";
    cin >> n;

    for(int i=1; i<=n; i++) {
        cout << i << " ";
    }
    cout << endl;

    return 0;
}
```


PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
```



Print sum of first N natural numbers

$$\underbrace{1 + 2 + 3 + 4 + 5 \ldots\ldots + N}_{\text{sum}}$$

1 to N

Cpp Codes

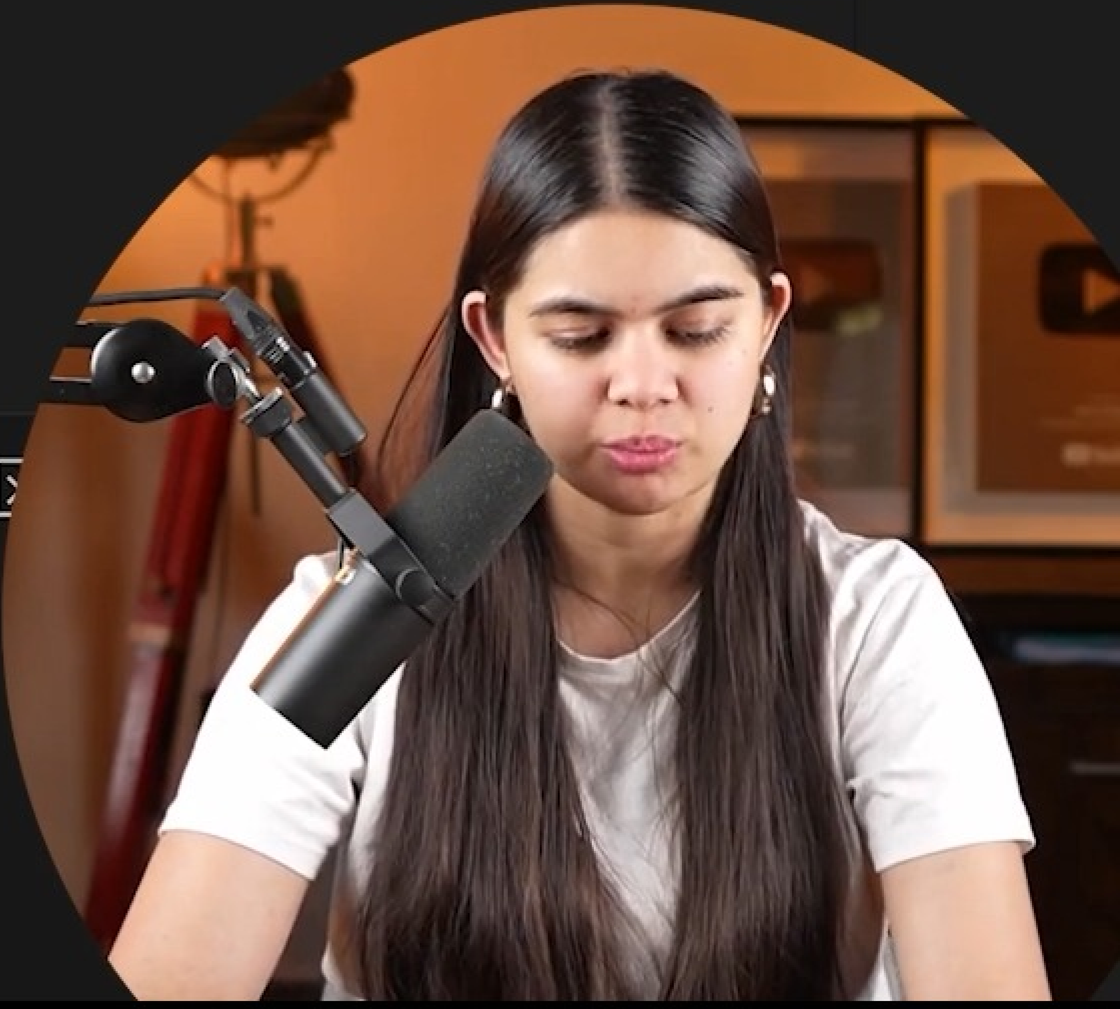
code.cpp

code.cpp > main()

```cpp
int main() {
    int n;
    cout << "enter your n : ";
    cin >> n;

    int sum = 0;
    for(int i=1; i<=n; i++) {
        sum += i;
    }
    cout << "sum = " << sum << endl;

    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
enter your n : 3
sum = 6
apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
enter your n : 5
sum = 15
apnacollege@Amans-MacBook-Pro Cpp Codes %
```



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Print sum of first N natural numbers

$$\underbrace{1 + 2 + 3 + 4 + 5 \; \ldots\ldots \; + N}_{\text{sum}}$$

while Loop

Loop runs while condition is true

```
int count = 1; //init
while(count < 3) {
    cout << count;
    count++; //updation
}
```

Cpp Codes

code.cpp

code.cpp > main()

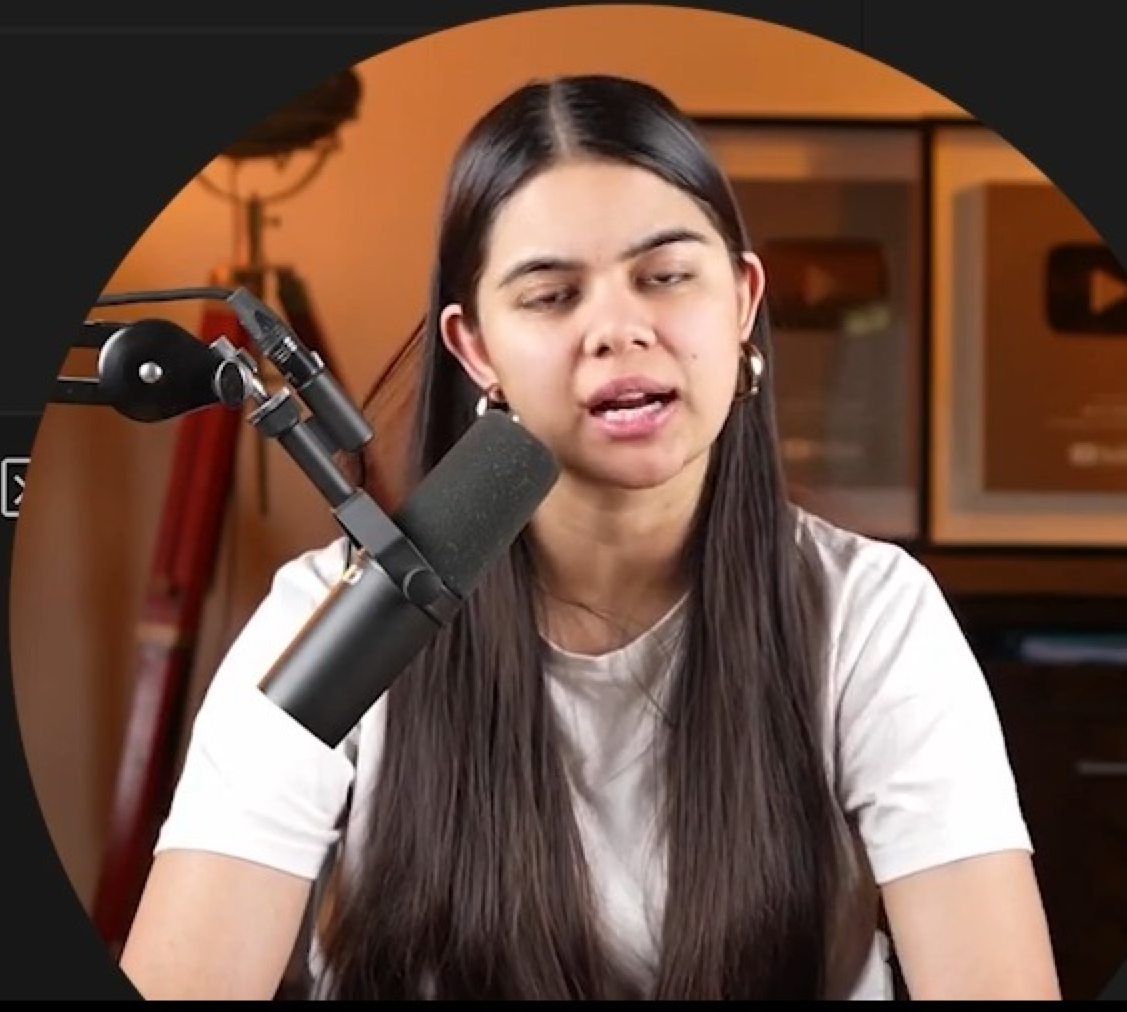
```cpp
#include <iostream>
using namespace std;

int main() {
    int count = 1;
    while(count < 3) {
        cout << count << " ";
        count++;
    }

    cout << endl;
    return 0;
}
```

PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

```
apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
1 2
apnacollege@Amans-MacBook-Pro Cpp Codes %
```



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Practice Qs

Qs. Print the Square pattern using ***for*** loop.

```
1 → * * * *
2 → * * * *
3 → * * * *
4 → * * * *
```

Qs. Print numbers from n to 1 using ***for*** loop.

Cpp Codes

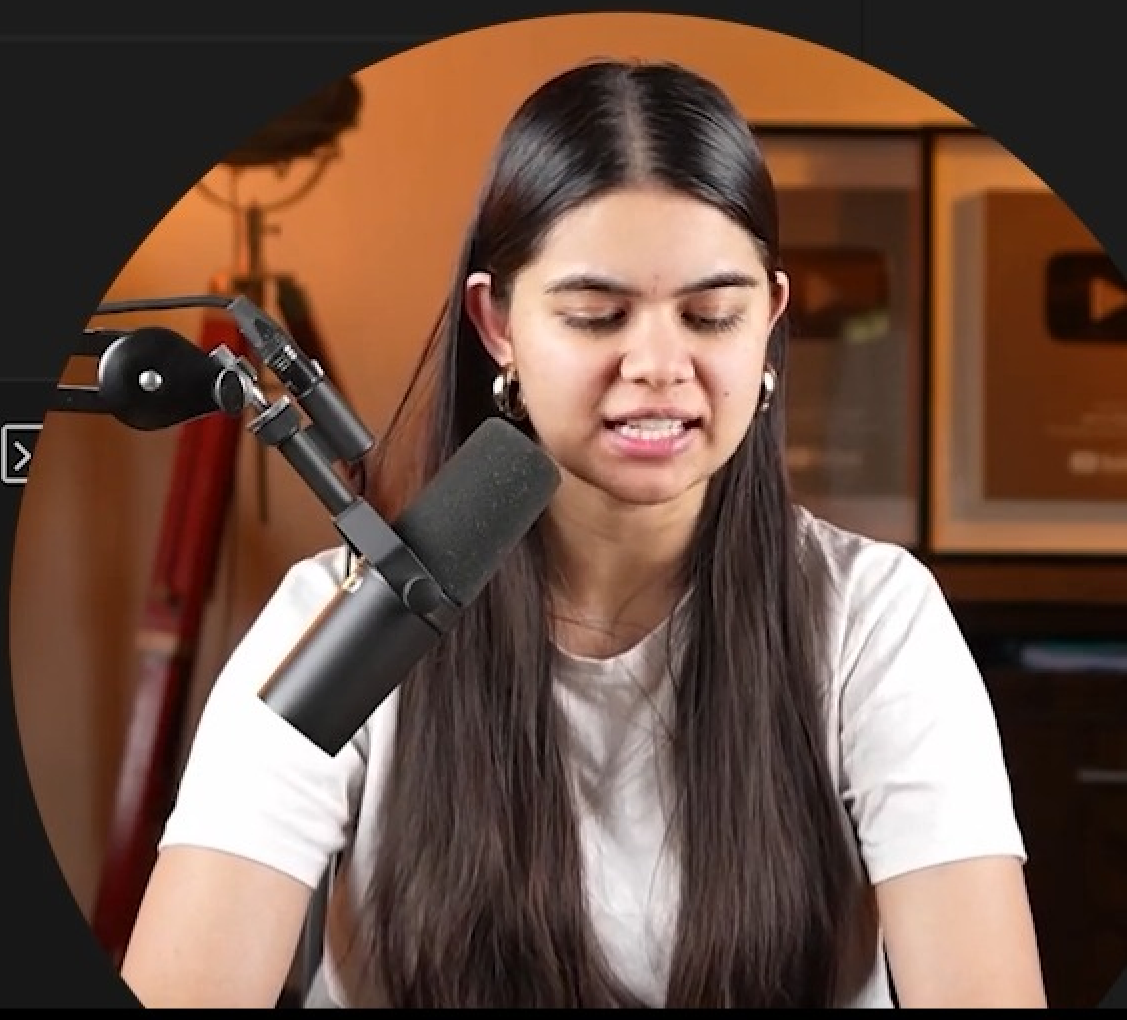
code.cpp

code.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main() {
    int n = 12;

    for(int i=n; i>=1; i--) {
        cout << i << " ";
    }

    cout << endl;
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
12 11 10 9 8 7 6 5 4 3 2 1
apnacollege@Amans-MacBook-Pro Cpp Codes %
```



last Digit access

Practice Qs

number % 10 → last Digit

Qs. Print the sum of digits of a number using ***while*** loop.

n = 10829

1 + 0 + 8 + 2 + 9

17 % 10 → 7

256 % 10 → 6

Qs. Print the sum of odd digits of a number using ***while*** loop.

Practice Qs

Qs. Print the sum of digits of a number using ***while*** loop.

n = 10829

$17 / 10 \to 1$

$235 / 10 \to 23$

lastDig = n % 10

sum += lastDig

23

10) 235

20

35

−30

5

Qs. Print the sum of odd digits of a number using ***while*** loop.

Practice Qs

$\boxed{lastDig = n \% 10}$

$sum += lastDig$

Qs. Print the sum of digits of a number using ***while*** loop.

n = 10829 /10 = 1082 → 9

1082 /10 = 108 → 2

108 /10 = 10 → 8

10 /10 = 1 → 0

1 /10 = 0 1

Qs. Print the sum of odd digits of a number using ***while*** loop.

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Practice Qs

Qs. Print the sum of digits of a number using ***while*** loop.

n = 10829 /10 = 1082 → 9

1082 /10 = 108 → 2

108 /10 = 10 → 8

10 /10 = 1 → 0

1 /10 = 0   1

lastDig = n % 10

sum+ = lastDig

n = n / 10

Qs. Print the sum of odd digits of a number using ***while*** loop.

Practice Qs

Qs. Print the sum of digits of a number using ***while*** loop.

n = 10829

Qs. Print the sum of odd digits of a number using ***while*** loop.

1 0 8 2 9

10

if (lastDig % 2 != 0)

Practice Qs

Qs. Print the sum of digits of a number using ***while*** loop.

n = 10829

Qs. Print the sum of odd digits of a number using ***while*** loop.

1 0 8 2 9

10

if (lastDig % 2 != 0)

sum += lastDig

Practice Qs

Qs. Print the digits of a given number in reverse using ***while*** loop.

n = 10829 → 9 2 8 0 1

```
n

lastDig = n % 10
cout<<lastDig
n = n/10
```

Qs. Reverse a given number & print the result.

Cpp Codes

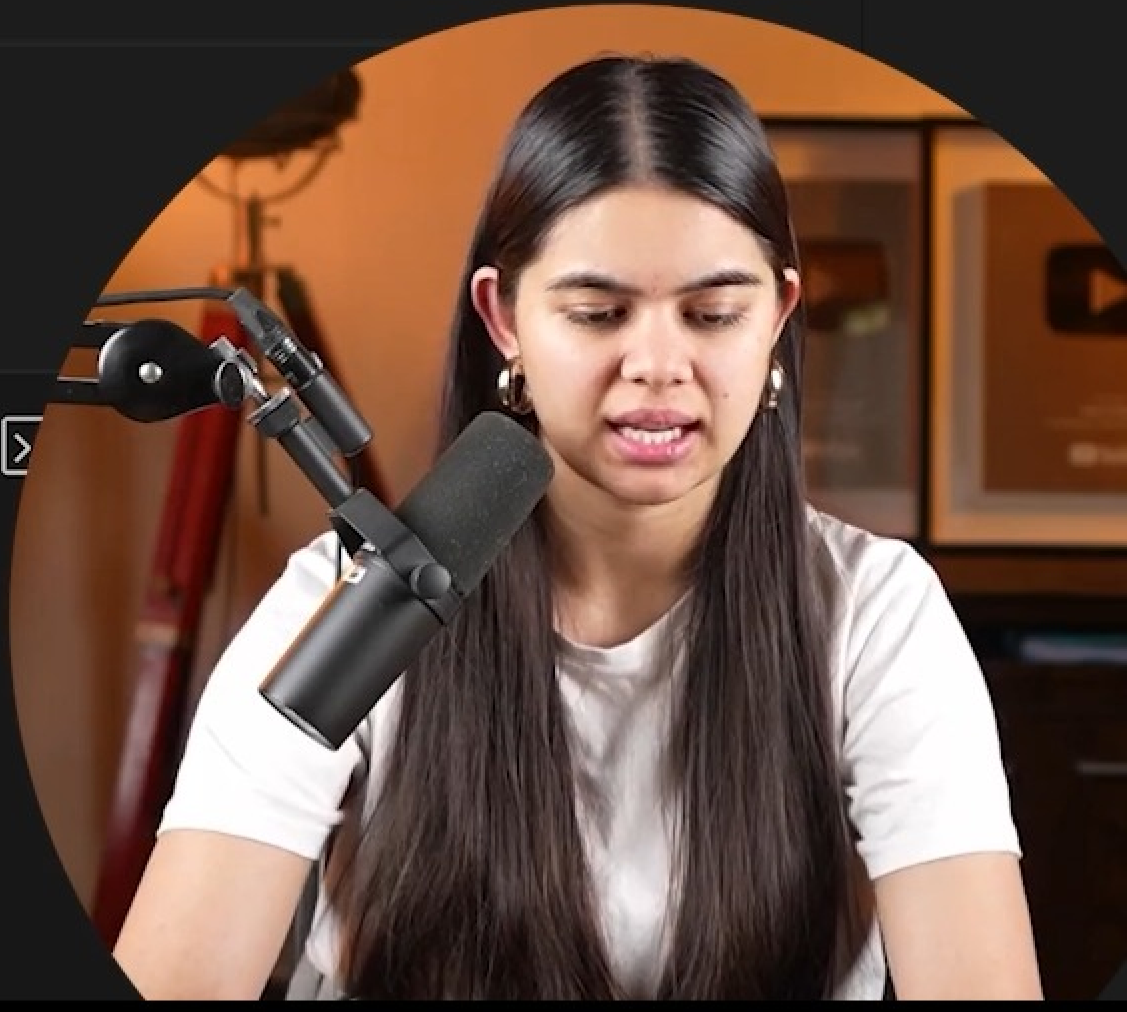
code.cpp

code.cpp > main()

```cpp

int main() {
    int n = 10829;

    while(n > 0) {
        int lastDig = n % 10;
        cout << lastDig;
        n /= 10;
    }

    cout << endl;
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
apnacollege@Amans-MacBook-Pro Cpp Codes %
```



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Practice Qs

Qs. Print the digits of a given number in reverse using ***while*** loop.

n = 10829

last Dig = n % 10

$\boxed{res = res * 10 + last Dig}$ ✓

n = n / 10

5 2 3

$5 \times 10^2 = 500$

$+ \; 2 \times 10^1 = 20$

$+ \; 3 \times 10^0 = 3$

523

Qs. Reverse a given number & print the result.

res = 0 $\quad 0 * 10 + 9 = 9$

res = 9 $\quad 9 * 10 + 2 = 92$

res = 92 $\quad 92 * 10 + 8 = 928$

res = 928 $\quad 928 * 10 + 0 = 9280$

res = 9280 $\quad 9280 * 10 + 1 = 92801$

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Cpp Codes

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code.cpp

code.cpp > main()

```cpp
int main() {
    int n = 10829;
    int res = 0;

    while(n > 0) {
        int lastDig = n % 10;
        res = res * 10 + lastDig;
        n /= 10;
    }

    cout << "reverse = " << res << endl;
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
reverse = 54321
apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
reverse = 92801
apnacollege@Amans-MacBook-Pro Cpp Codes %
```

do-while Loop

```
do {

    //do some work

} while(condition);
```

//work gets done atleast once irrespective of condition

do-while Loop

```
do {

    //do some work

} while(condition);
```

//work gets done atleast once irrespective of condition

while

```
while ( ){

}
```

do-while

```
do {

} while( );
```

Cpp Codes

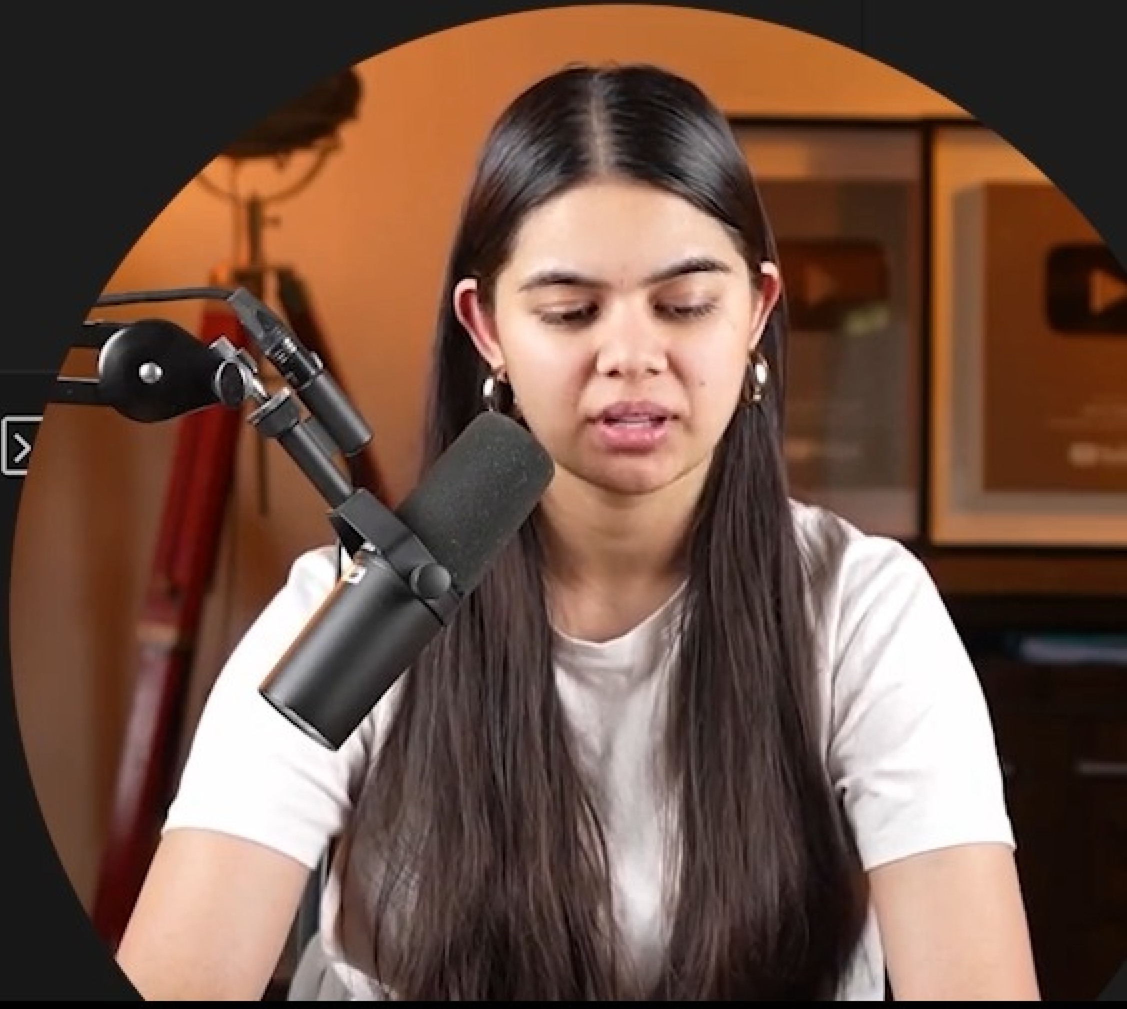
code.cpp

code.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main() {
    int i = 1;

    do {
        cout << i << " ";
        i++;
    } while(i <= 5);

    cout << endl;
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
1 2 3 4 5
apnacollege@Amans-MacBook-Pro Cpp Codes %
```



Break Statement

to exit the loop

Cpp Codes

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code.cpp ×

code.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main() {
    int i = 1;
    while(i <= 10) {
        if(i == 3) {
            break;
        }
        cout << i << endl;
        i++;
    }

    cout << "out of loop now" << endl;
    return 0;
}
```

PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

```
apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
1
2
out of loop now
apnacollege@Amans-MacBook-Pro Cpp Codes % 
```

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Practice Qs

Qs. WAP where user can keep entering numbers till they enter a multiple of 10.

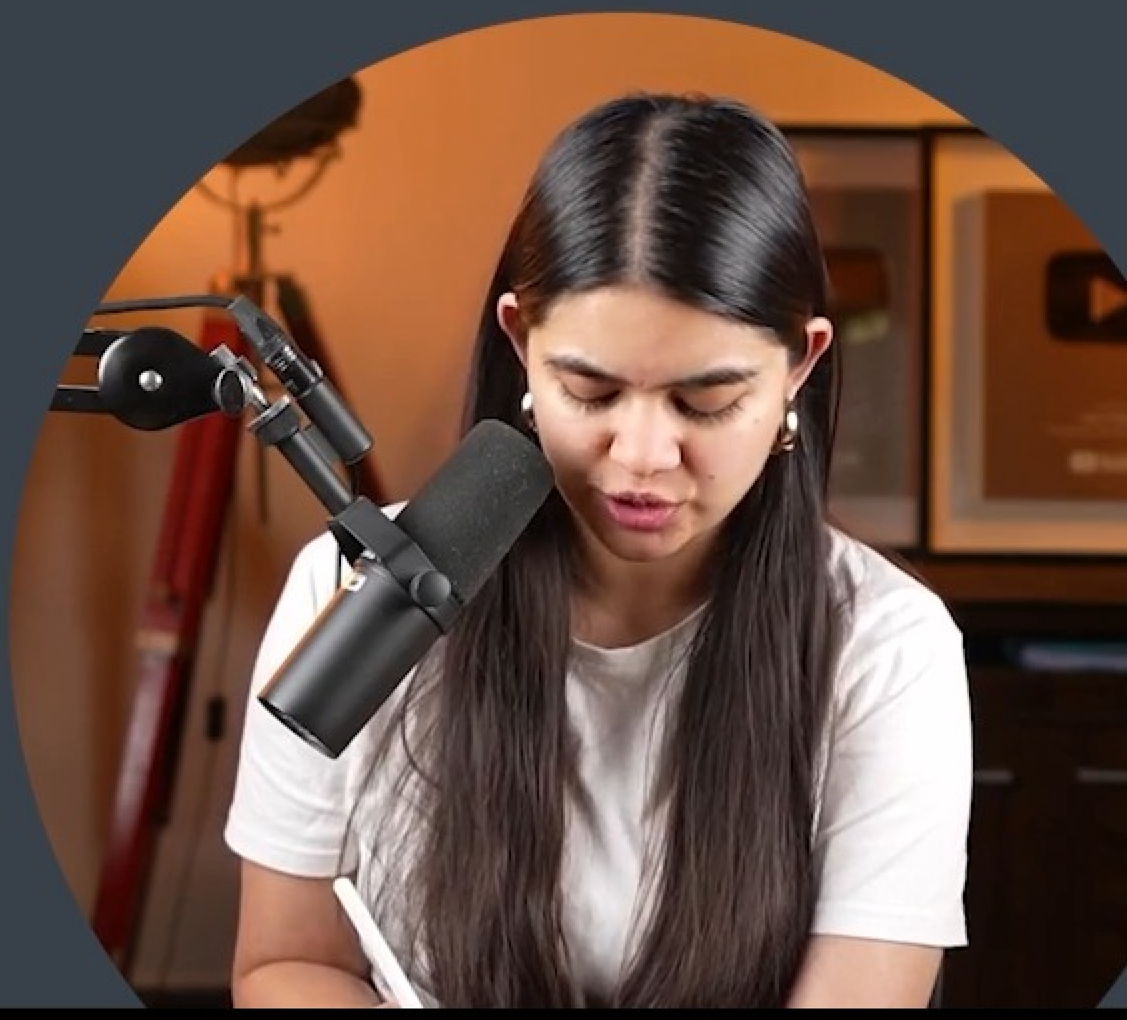
5

15

37

64

50 X loo



Cpp Codes

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code.cpp

code.cpp > main()

```cpp
using namespace std;

int main() {
    int n;

    do {
        cout << "enter number : ";
        cin >> n;
        if(n % 10 == 0) {
            break;
        }
        cout << "you entered " << n << endl;
    } while(true);

    return 0;
}
```

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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
1
2
out of loop now
apnacollege@Amans-MacBook-Pro Cpp Codes %
```

Continue Statement

to skip an iteration

Practice Qs

Qs. WAP to show numbers entered by user except multiples of 10.

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Cpp Codes

code.cpp

code.cpp > main()

```cpp
using namespace std;

int main() {
    int n;

    do {
        cout << "enter number : ";
        cin >> n;
        if(n % 10 == 0) {
            continue;
        }
        cout << "you entered " << n << endl;
    } while(true);

    return 0;
}
```

Practice Qs

Qs. Check if a number is **prime** or not.

$n = 2, 3, 5, 7, 11, 13 \ldots$

$5 \Rightarrow 1 \times 5$
$(1, 5)$ 5×1

$7 \Rightarrow 1 \times 7$
$(1, 7)$ 7×1

$17 \Rightarrow 1 \times 17$
$(1, 17)$ 17×1

$6 \Rightarrow 1 \times 6$
2×3
3×2
6×1

$8 \Rightarrow 1 \times 8$
2×4
4×2
8×1

$12 \Rightarrow 1 \times 12$
2×6
3×4
4×3
6×2
12×1

$12 \% 4 \Rightarrow 0$

$x \; (1x, nx)$

$[2, n-1]$

num $\Rightarrow$ composite

$\underline{\text{num} \% x = 0}$

Practice Qs

Qs. Check if a number is **prime** or not.

$n = 2, 3, 5, 7, 11, 13 \ldots$

```
for( int x = 2; x <= n-1 ; x++ ){
```

$n \% x \to 0$ composite

$n \% x \to \times$ prime

```
}
```

$12 \Rightarrow 1 \times 12$, 2×6, 3×4, 4×3, 6×2, 12×1

$12 \% 4 \Rightarrow 0$

$x \; (1x, nx)$ $\quad [2, n-1]$

$num \% x = 0$

$num \Rightarrow$ composite

Practice Qs

Qs. Check if a number is **prime** or not.

$3^2 \Rightarrow 9$

$4^2 \Rightarrow 16$

$12 \Rightarrow$

1×12

2×6

3×4

4×3

6×2

12×1

inc

dec

$\sqrt{12} \rightarrow 3. =$

$12 \% 4 \Rightarrow 0$

$15 \rightarrow$

1×15

3×5

5×3

15×1

$\sqrt{15}$

$3.9-$

$a * b$ maximum

$\sqrt{n} * \sqrt{n} = n$

Practice Qs

Qs. Check if a number is **prime** or not.

$21 \Rightarrow$

1×21

3×7

7×3

21×1

$\sqrt{21} \Rightarrow 4.\_$

$\underbrace{2, 3, 4}$

$\underbrace{2 \text{ to } \sqrt{n}}$

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code.cpp

code.cpp > main()

```cpp
#include <iostream>
#include <cmath>
using namespace std;

int main() {
    int n = 21;
    bool isPrime = true;

    for(int i=2; i<=sqrt(n); i++) {
        if(n % i == 0) { // i is a factor of n; i completely divides n; n is non-prime
            isPrime = false;
            break;
        }
    }
```

PORTS PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
number is NOT Prime
apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
number is NOT Prime
apnacollege@Amans-MacBook-Pro Cpp Codes %
```